

Question 1:

Find FIVE mistakes in the proof.

Each mistake should correspond to either

- A violation of the *Directions for Writing Proofs*, OR
- One of the listed *Common Mistakes*.

Theorem

If n is an even integer, then $n^2 + n$ is even.

Proof:

Let $n = 2k$ be any integer. For example, if $n = 2$, then $n^2 + n = 4 + 2 = 6$ which is even.

Now since n is even, we can write $n = 2k$, Then

$$n^2 + n = (2k)^2 + 2k = 4k^2 + 2k = 2(2k^2 + k)$$

Since k is even, $2k^2 + k$ must be even. Therefore $n^2 + n$ is divisible by 2.

Thus the statement is true for any integer.

SOLUTION:

1. Arguing from an example: The proof uses $n = 2$ as an example, which does not prove a universal statement.
2. Violates Writing Direction (complete, grammatically correct sentences): 'Let $n = 2k$ be any integer' is wrong wording; it should say 'Let n be an even integer, then $n = 2k$ for some integer k .'
3. Using the same letter improperly / unclear logic: The proof repeats ' $n = 2k$ ' without clearly stating k is an integer.
4. False assumption: It incorrectly states 'Since k is even', which is not given and cannot be concluded.
5. Incorrect conclusion: It says the statement is true for any integer, while the theorem is only about even integers.

Question 2:

Prove the statement and give the counterexample if it is false. For all integers a , b and c if $a \mid b$ and $a \mid c$ then $a \mid 2b - 3c$.

SOLUTION:

Proof:

Assume $a \mid b$ and $a \mid c$.

Then there exist integers k and m such that $b = ak$ and $c = am$.

$$2b - 3c = 2(ak) - 3(am).$$

$$2b - 3c = a(2k - 3m).$$

Since $(2k - 3m)$ is an integer, it follows that $a \mid (2b - 3c)$.

Therefore, the statement is TRUE, so no counterexample exists.